



JP Hwang

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Weaviate





Powering Multimodal Search and RAG with Weaviate



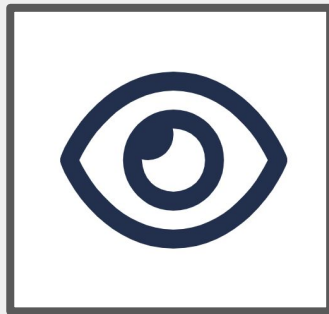
JP Hwang
Developer Educator

Multimodality

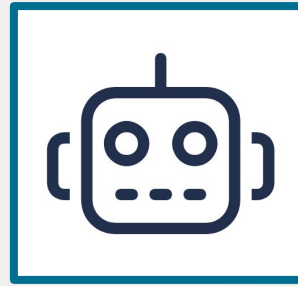




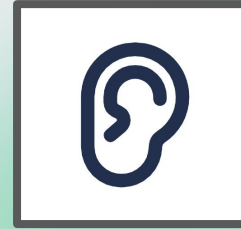
We are
Multimodal



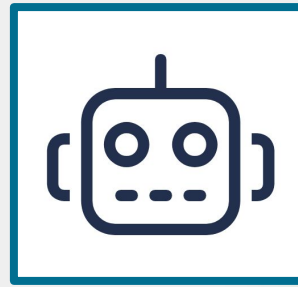
AI & Multimodality



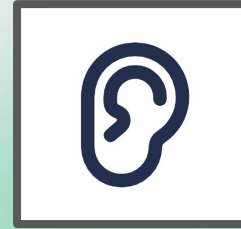
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AI & Multimodality **Why?**



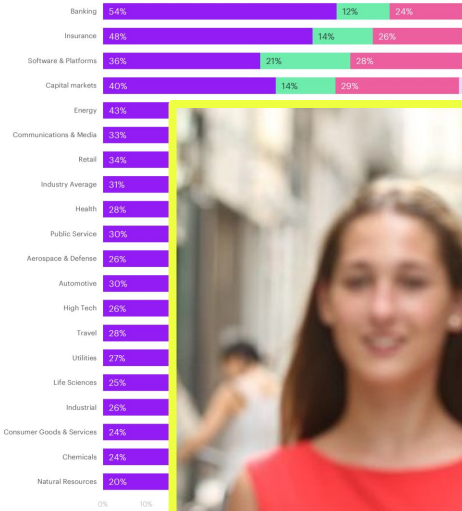
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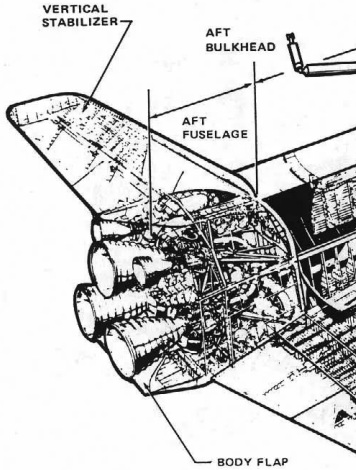


AI & Multimodality: **Why?**

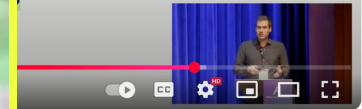
Figure 3: Generative AI will transform work across industries



Activating and Deactivating Tenants

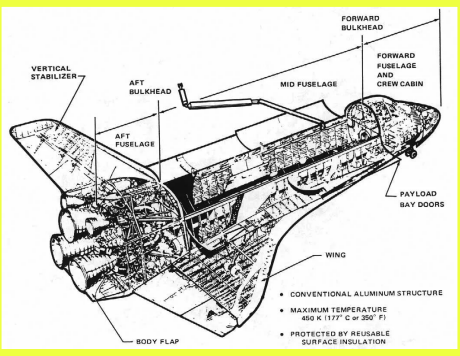


- MAXIMUM 450 K (1
- PROTECTE SURFAC



ne Dilocker, CTO, Weaviate

Share Download Clip



Activating and Deactivating Tenants

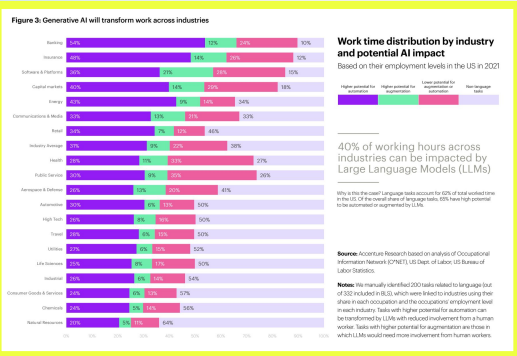
Active Tenants
are already in memory
and **ready to go**

Inactive Tenants
are on disk, but can be loaded
in a **few hundred ms**

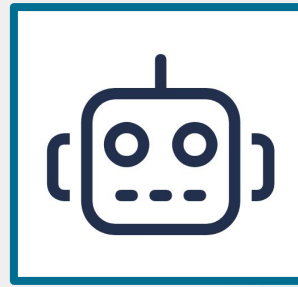
Solving Multi-Tenancy In Vector Search Requires A Paradigm Shift, Etienne Dilorcker, CTO, Weaviate

The AI Conference
1.18K subscribers

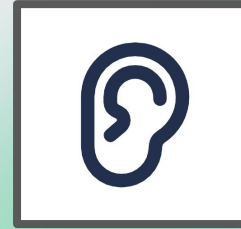
40 Share Download Clip



AI & Multimodality **Why?**

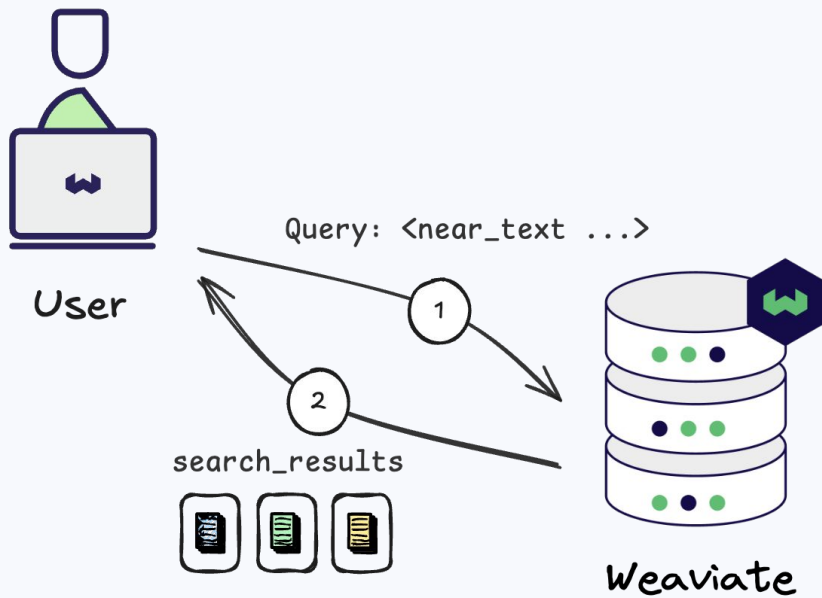


+



RAG

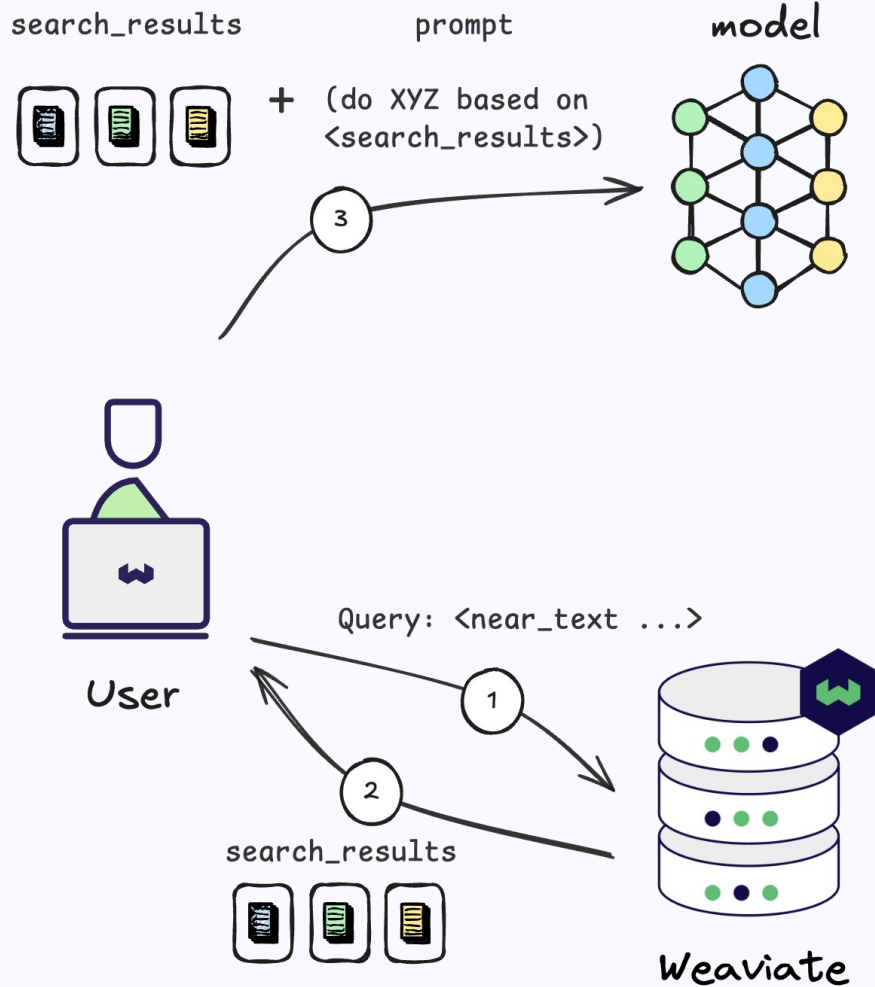
(Retrieval
augmented
generation)





RAG

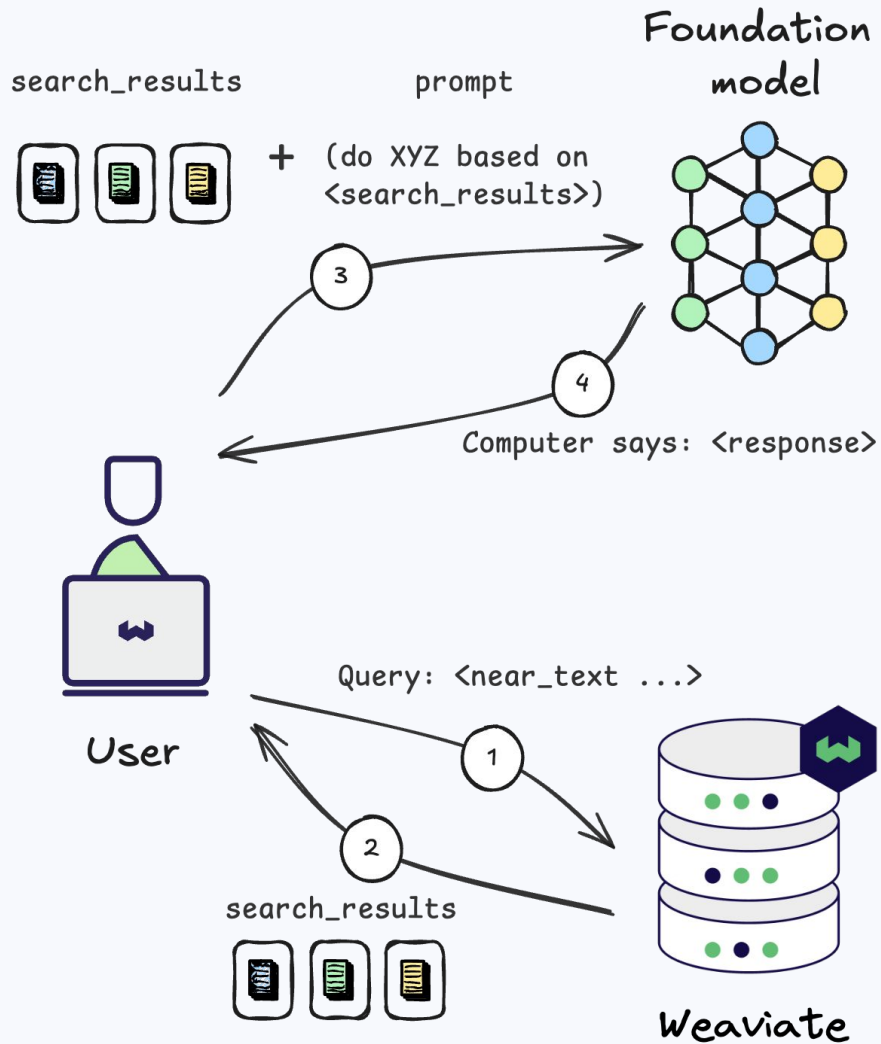
(Retrieval
augmented
generation)





RAG

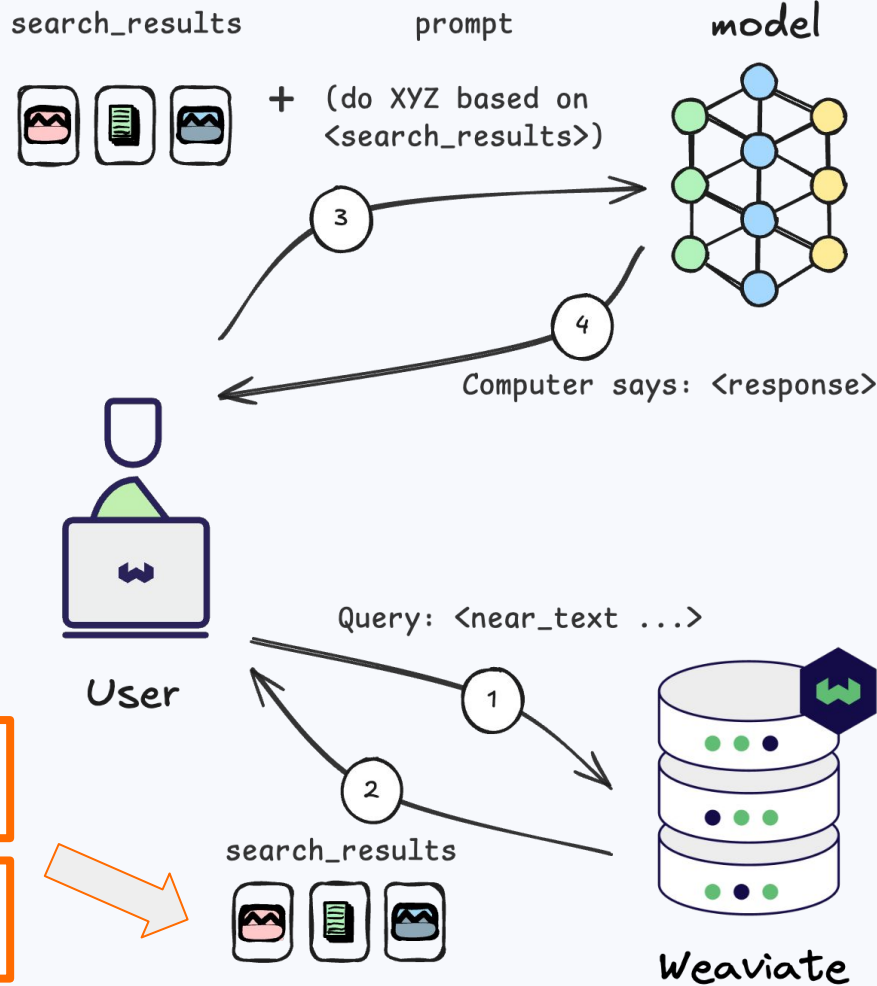
(Retrieval
augmented
generation)





RAG

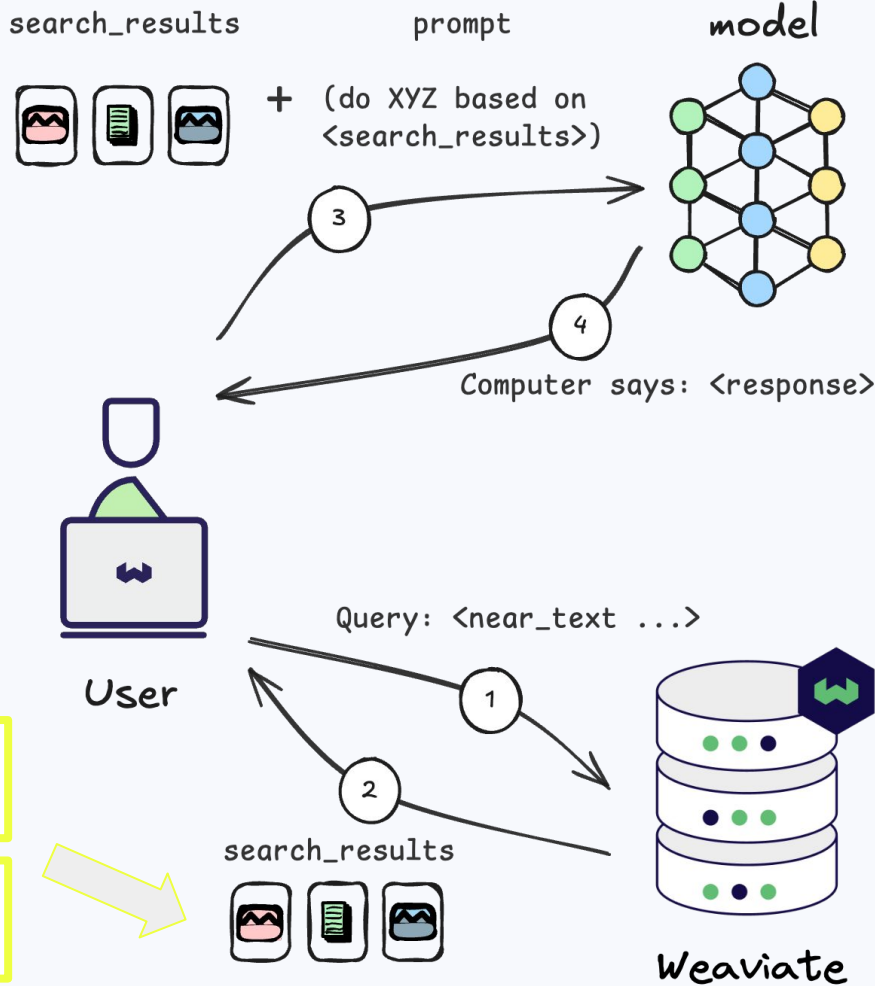
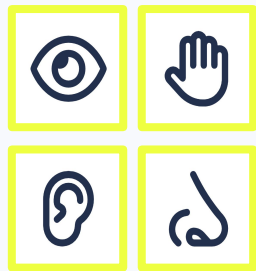
(Retrieval
augmented
generation)





RAG

(Retrieval
augmented
generation)





A Quick Demo

 [Demo app](#)

Types of multi-modal models

Generative

- Text / Images → Images
- Text / Images → Text
- Diffusion / Dall-E
- VLMs (Claude 3 / GPT-4 / Gemini)

Embeddings

- Text / Images → Embeddings
- CLIP / SigLIP / ColPali



How Multimodal Models Work

What does this sound like?

We can infer information from another modality



Multimodal Reasoning and Retrieval

Understand with all our senses

Multimodal understanding



Multimodal Reasoning and Retrieval

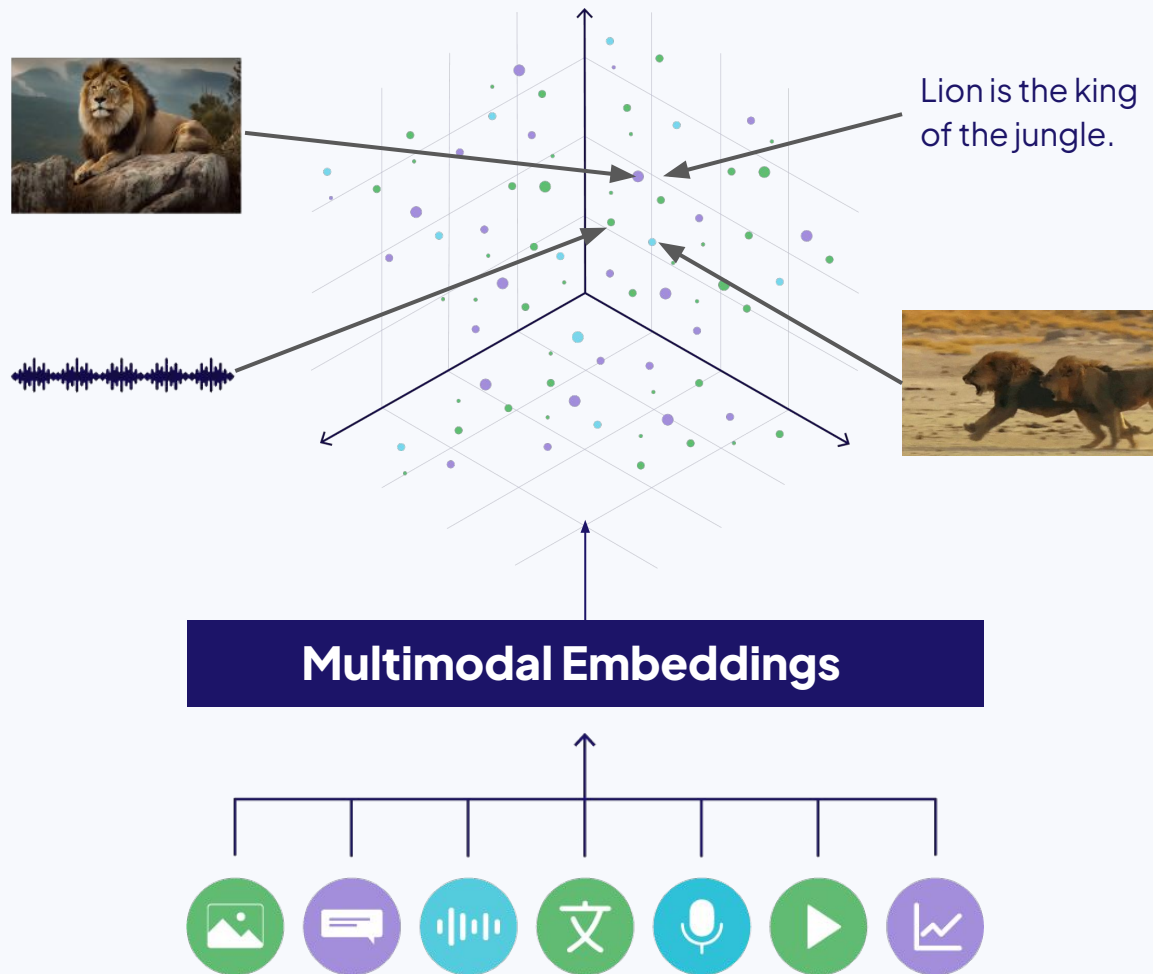
Multimodal reasoning works in all directions

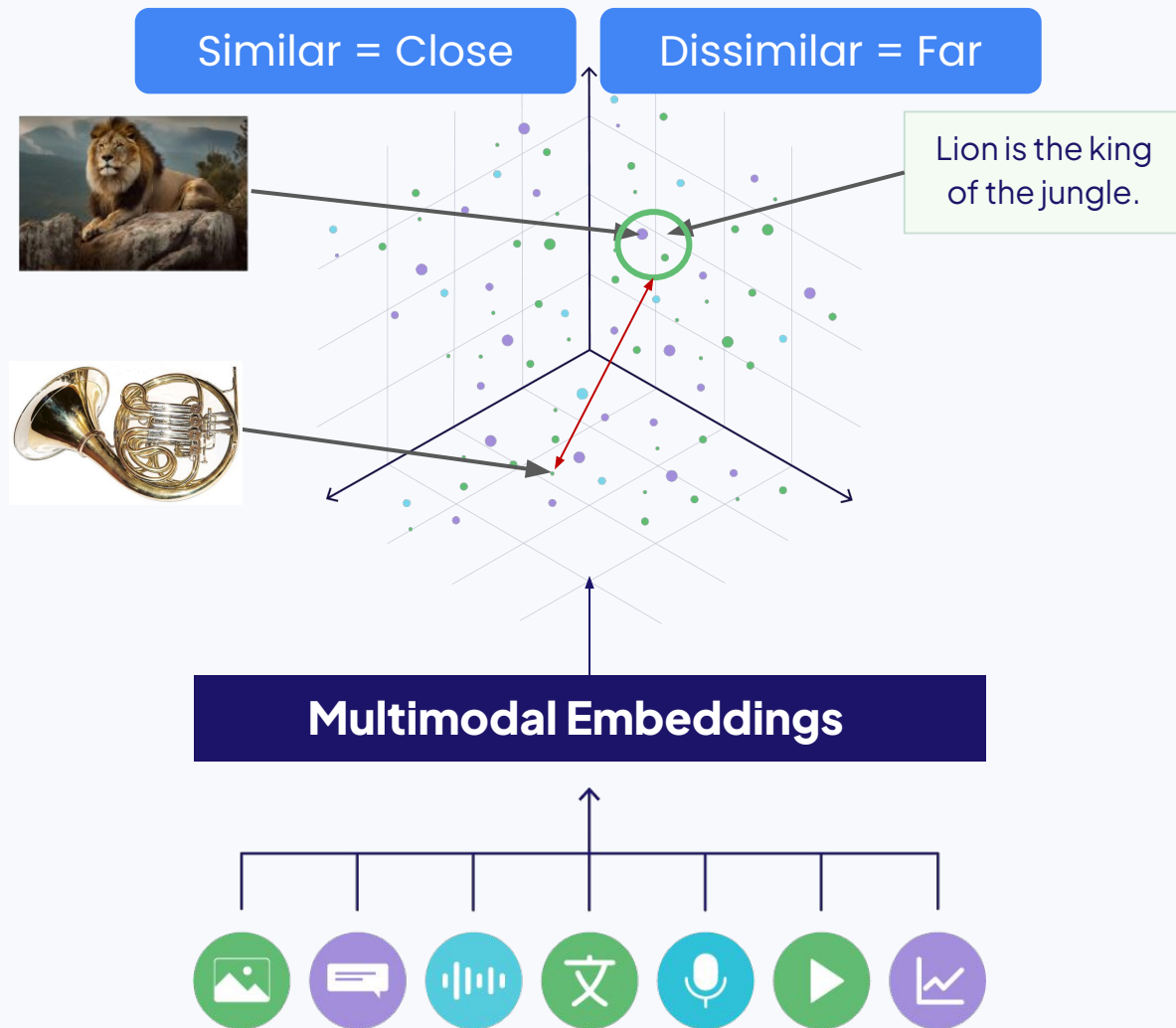




Generally Capable AI needs to be multimodal.







Start with specialist models

The lion is the king
of the savannas.



Text Encoder

$[-0.10, 0.02, 0.14, \dots, 0.48]$



Image Encoder

$[0.66, 0.73, -0.42, \dots, 0.52]$



Audio Encoder

$[-0.61, 0.31, 0.88, \dots, 0.22]$



Video Encoder

$[-0.21, 0.15, 0.27, \dots, -0.59]$



The lion is the king
of the savannas.



Text Encoder



Image Encoder



Audio Encoder



Video Encoder

$[-0.10, 0.02, 0.14, \dots, 0.43]$

$[0.61, 0.03, 0.04, \dots, 0.52]$

$[0.01, 0.03, 0.08, \dots, 0.25]$

$[0.21, 0.05, 0.02, \dots, -0.59]$

similar concepts = similar vectors

Anchor:

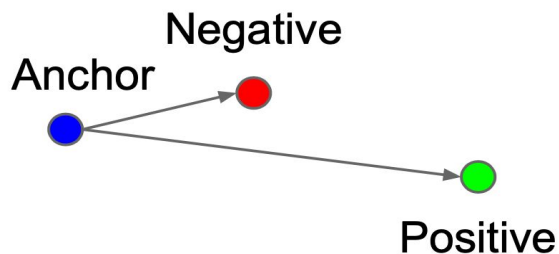
"He could smell the roses."

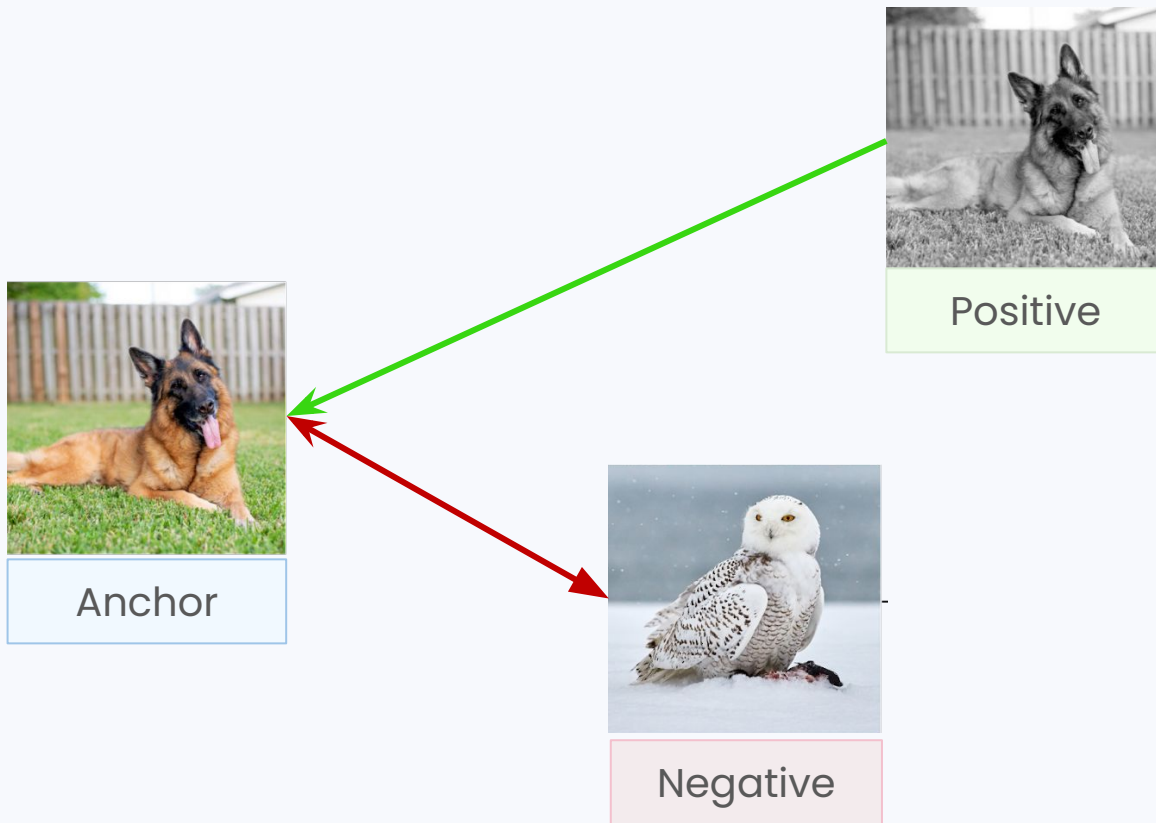
Negative Example:

"The lion roared majestically."

Positive Example:

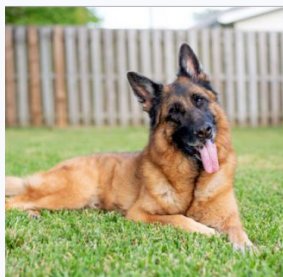
"A field of fragrant flowers."





Source: [V7 Labs](#)

Push Negative Example



Anchor



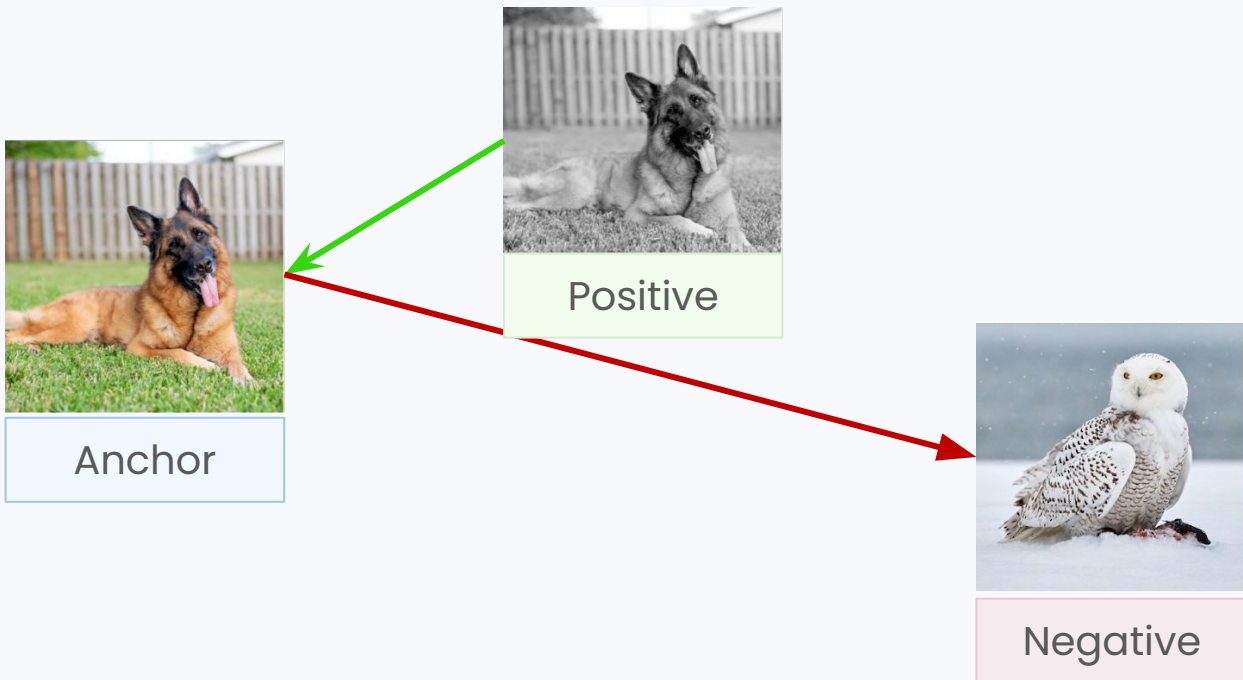
Positive



Negative

Source: [V7 Labs](#)

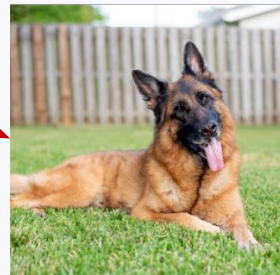
Pull Positive Example



Source: [V7 Labs](#)



**Lion is the king
of the jungle.**



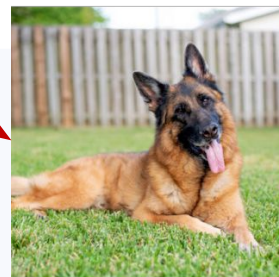
**Meerkats don't
consume water**



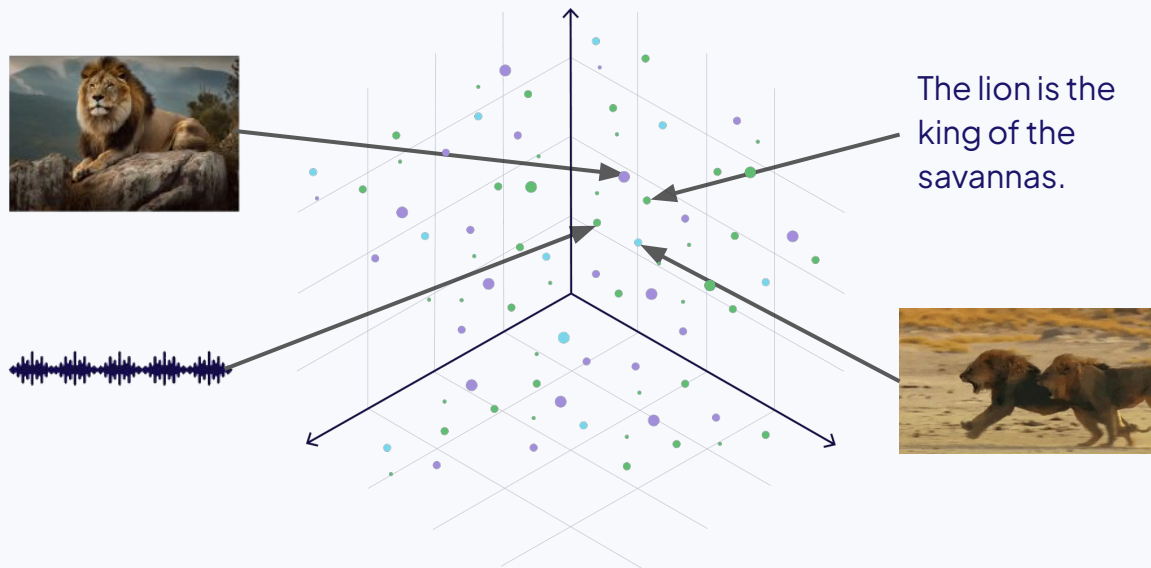
**Lion is the king
of the jungle.**



**Meerkats don't
consume water**



Shared Multimodal Vector Space

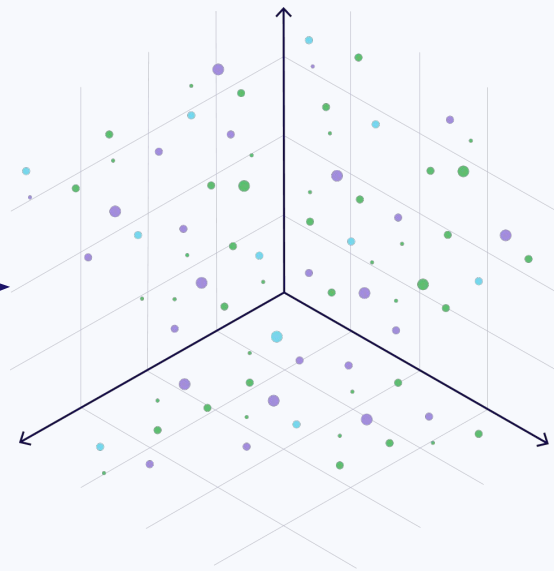


Text to Any Search

QUERY

RETRIEVED

Lions roam the
savannas



King of the
jungle.



Any to Any Search

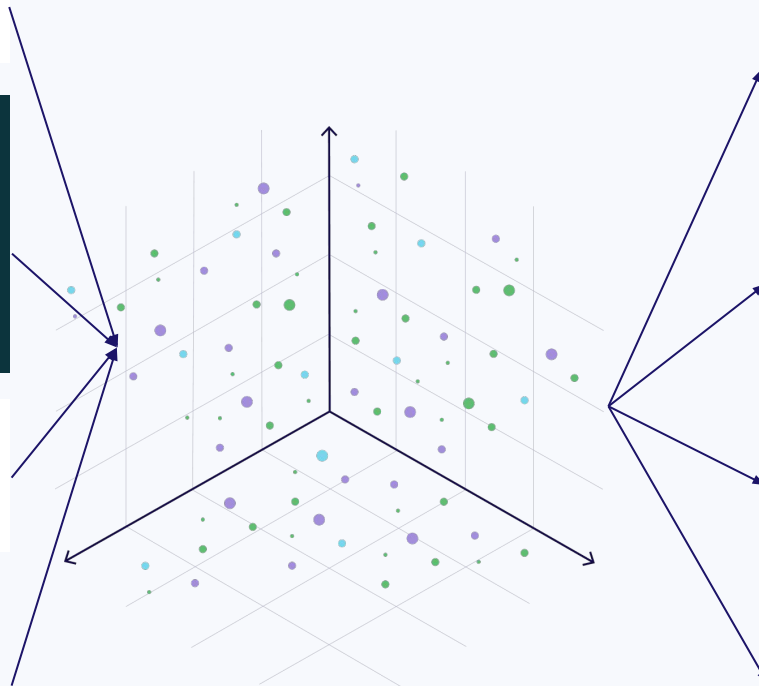
QUERY

RETRIEVED

Lions roam the savannas.



King of the jungle.



How do you teach LLMs to see?

Visual Instruction Tuning

Input: (Image, Text)

Output: Text



Who drew this
painting?

Vincent van Gogh

**Multimodal
models**

Me

**Unimodal
models**



AI & Multimodality: **Why?**

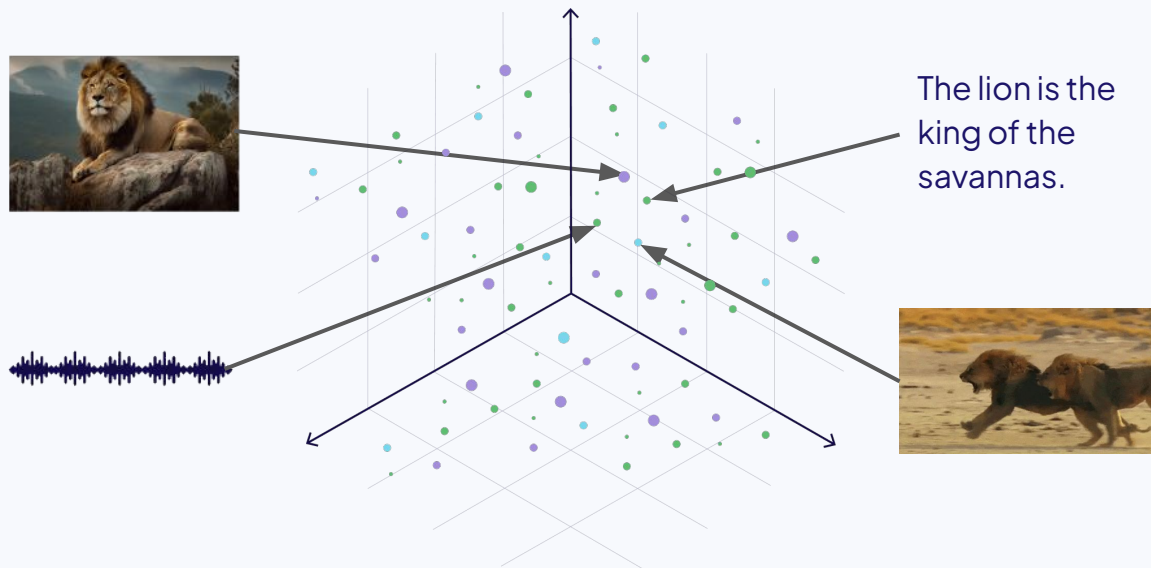
Multimodal RAG

- Feed images, videos, etc. as context to AI models
- Generate multimodal outputs
 - Images and text
 - Videos
 - Smells?
 - Actions?



Taking Multimodal RAG to *Production*

Shared Multimodal Vector Space



AI & Multimodality: Scalability

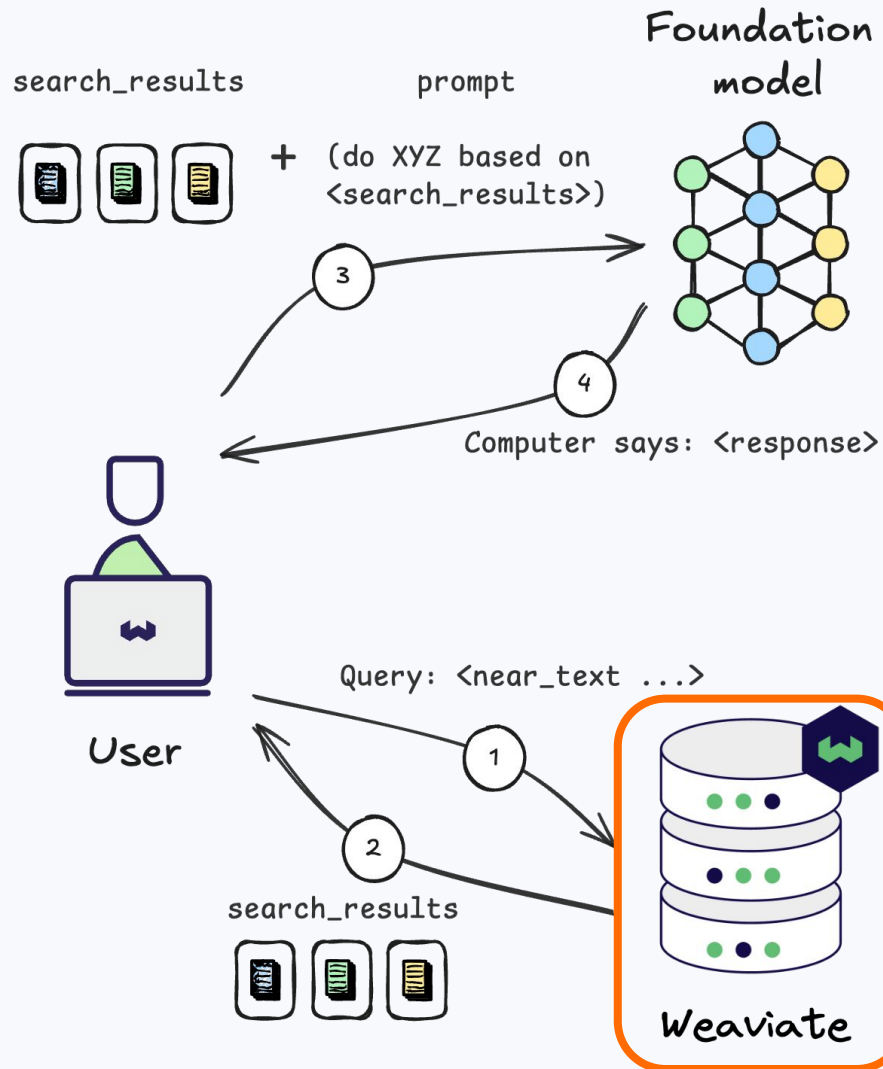
Scalability

- 100Ms of text vectors common (chunks)
- How many...
 - Photos (~2000/smartphone)
 - PDFs do you have?
 - Videos?

Search
is key

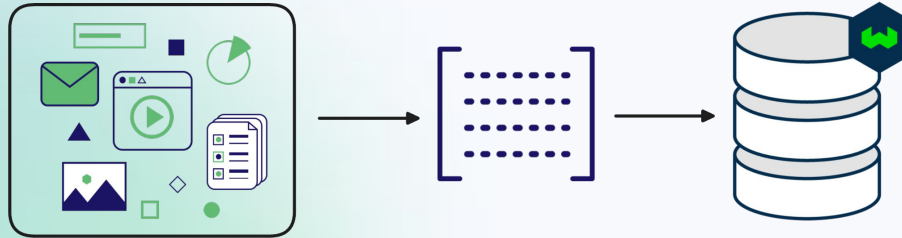


RAG



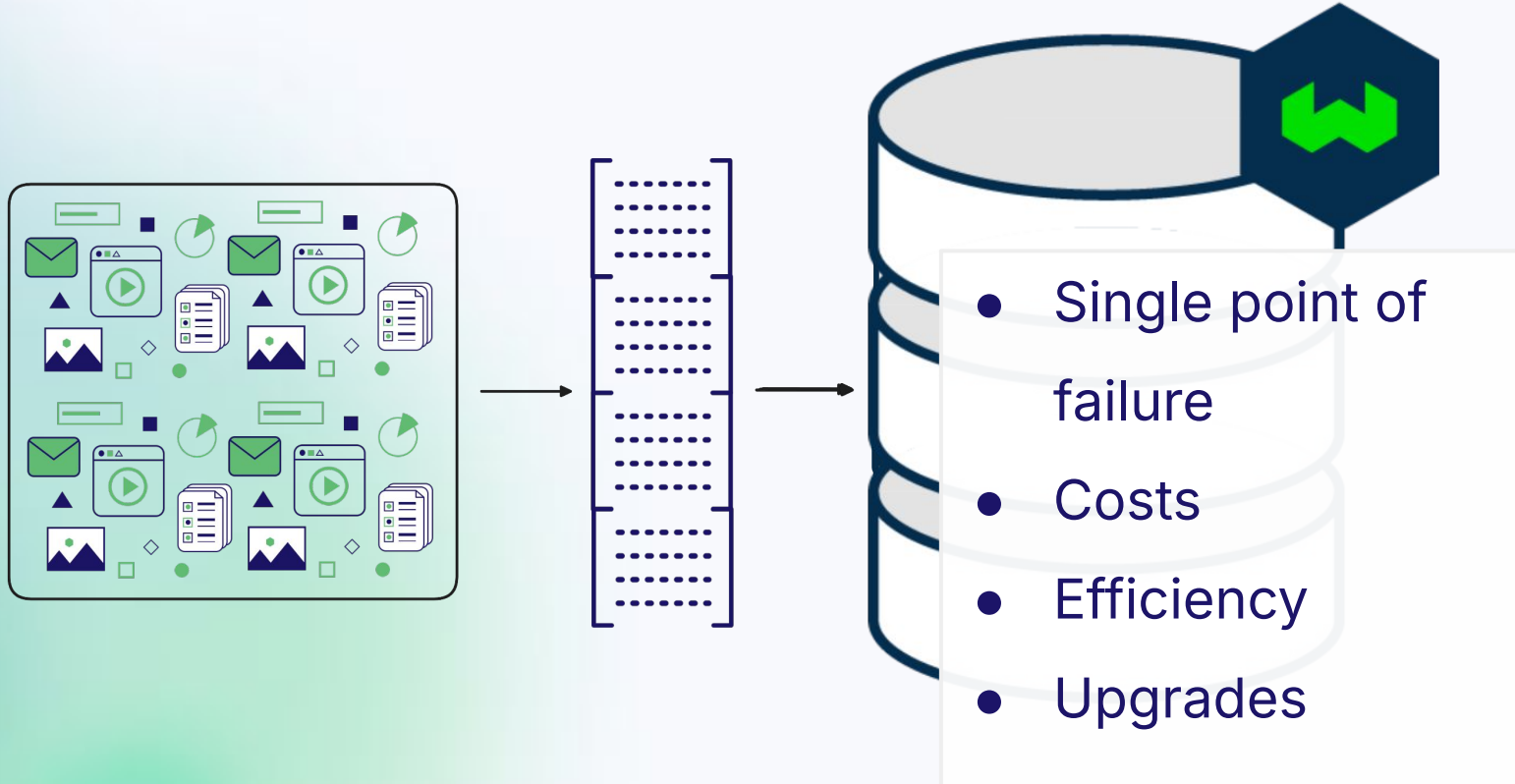
AI & Multimodality: Scalability

Scalability



AI & Multimodality: Scalability

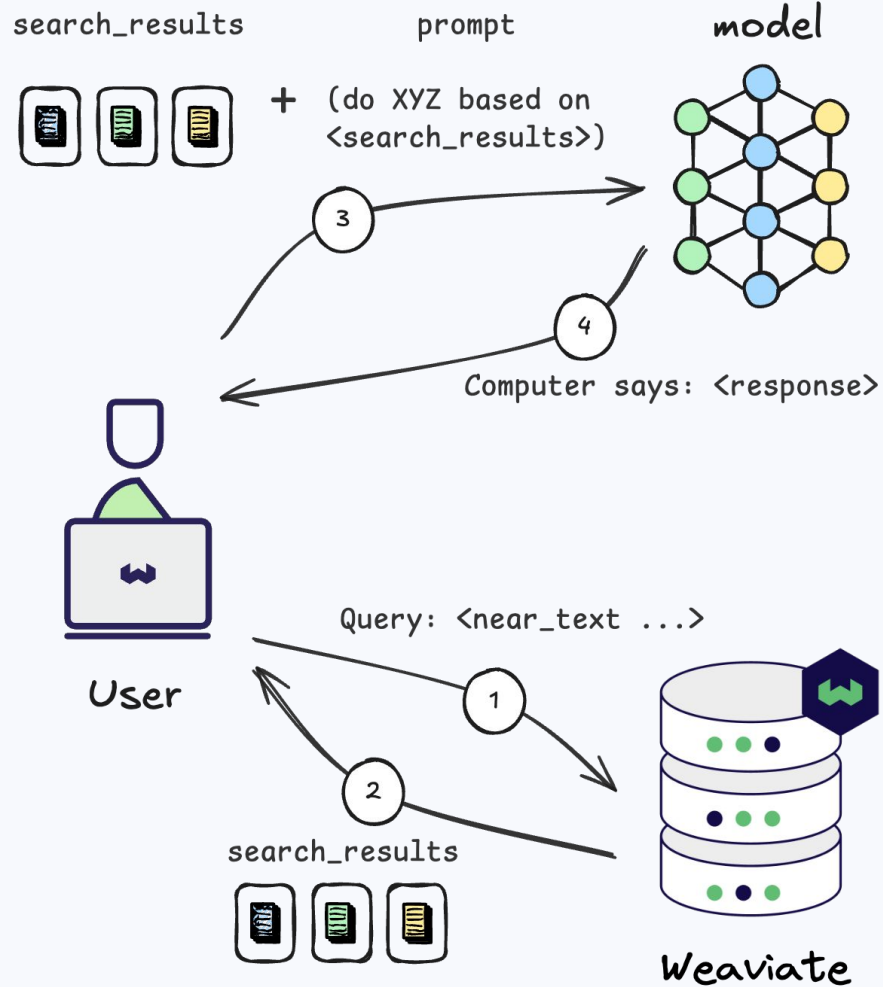
Scalability

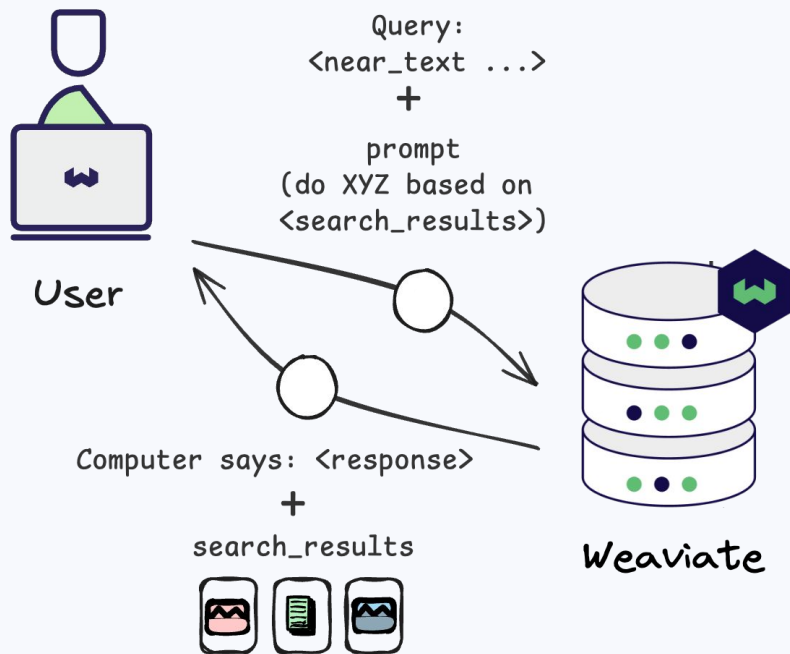


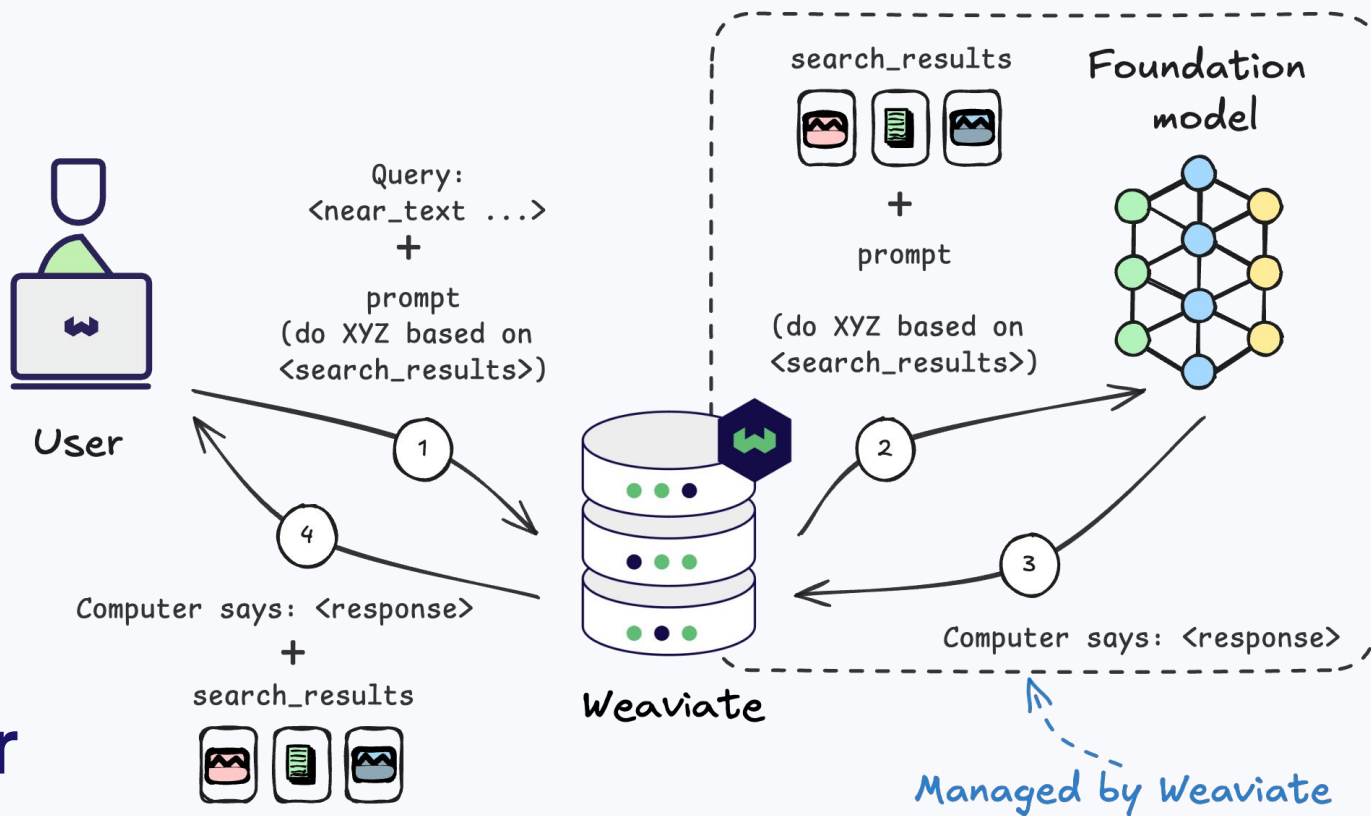
AI & Multimodality: **Scalability**

Scalability

- Scaling tools
 - Quantization
 - Multi-tenancy
 - Horizontal scaling
 - Replication







Developer Experience

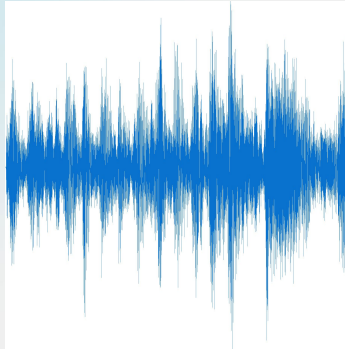


Application of MM-RAG

What burger do you like?



"Juicy, big, loaded with toppings of my choice. High quality beef medium to well with cheese and bacon on a multigrain bun. A huge single or triple burger with all the fixings, cheese, lettuce, tomato, onions and special sauce or mayonnaise!"



Nutrition Facts	
Valeur nutritive	
Per 1 burger (75 g) / pour 1 galette (75 g)	
Amount	% Daily Value
Teneur	% valeur quotidienne
Calories / Calories 100	
Fat / Lipides 2 g	3 %
Saturated / saturés 0.2 g	1 %
+ Trans / trans 0 g	
Cholesterol / Cholestérol 0 mg	
Sodium / Sodium 320 mg	13 %
Potassium / Potassium 400 mg	11 %
Carbohydrate / Glucides 7 g	2 %
Fibre / Fibres 2 g	8 %
Sugars / Sucres 1 g	
Protein / Protéines 13 g	



[0.65, 0.15, ..., 0.23, 0.75]

[0.23, 0.45, ..., 0.84, 0.23]

[0.03, 0.97, ..., 0.27, 0.26]



For RecSys MM representations allow us to:

- **More uniquely identify what customers like**
With MM we have more “senses” / modalities to do so
- **More uniquely compare relevance b/w products**
Compare across modalities
- **Identify differences amongst similar products**



(b) Query Product

Source <https://www.amazon.science/publications/unsupervised-multi-modal-representation-learning-for-high-quality-retrieval-of-similar-products-at-e-commerce-scale>

Que2Engage: Embedding-based Retrieval for Relevant and Engaging Products at Facebook Marketplace - <https://arxiv.org/pdf/2302.11052.pdf>



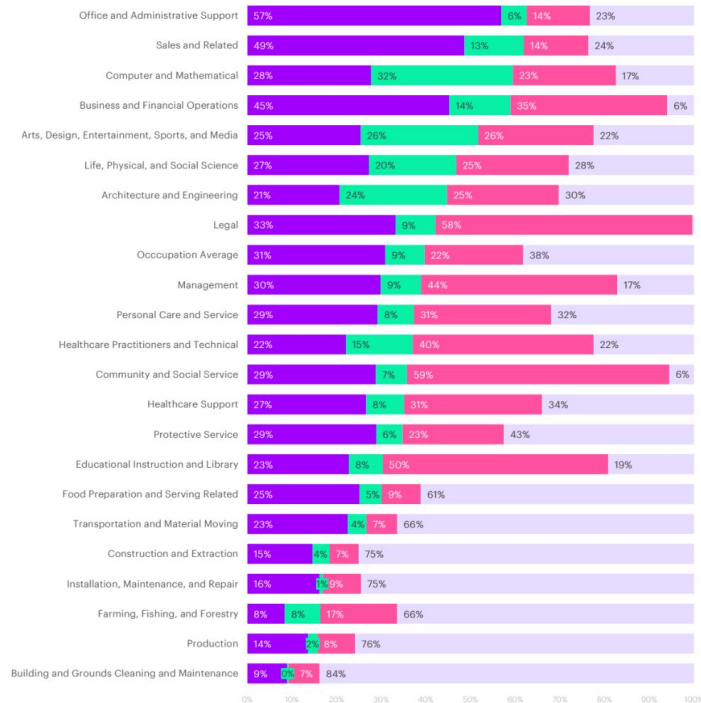
Some recent developments...

2

Take a people-first approach

Success with generative AI requires an equal attention on people and training as it does on technology. Companies should therefore dramatically ramp up investment in talent to address two distinct challenges: creating AI and using AI. This means both building talent in technical competencies like AI engineering and enterprise architecture and training people across the organization to work effectively with AI-infused processes. In our analysis across 22 job categories, for example, we found that LLMs will impact every category, ranging from 9% of a workday at the low end to 63% at the high end. More than half of working hours in 5 of the 22 occupations can be transformed by LLMs.

Figure 4: Generative AI will transform work across every job category



Work time distribution by major occupation and potential AI impact

Based on their employment levels in the US in 2021



In 5 out of 22 occupation groups, Generative AI can affect more than half of all hours worked

Source: Accenture Research based on analysis of Occupational Information Network (O*NET), US Dept. of Labor, US Bureau of Labor Statistics.

Notes: We manually identified 200 tasks related to language (out of 332 included in BLS), which were linked to industries using their share in each occupation and the occupations' employment level in each job category. Tasks with higher potential for automation can be transformed by LLMs with reduced involvement from a human worker. Tasks with higher potential for augmentation are those in which LLMs would need more involvement from human workers.



**USING COMPLEX RETRIEVAL SYSTEMS
THAT RELY ON OCR, DOCUMENT LAYOUT
RECOGNITION, CHUNKING STRATEGIES,
FIGURE CAPTIONING AND POWERFUL
TEXT EMBEDDING MODELS**

ColPali

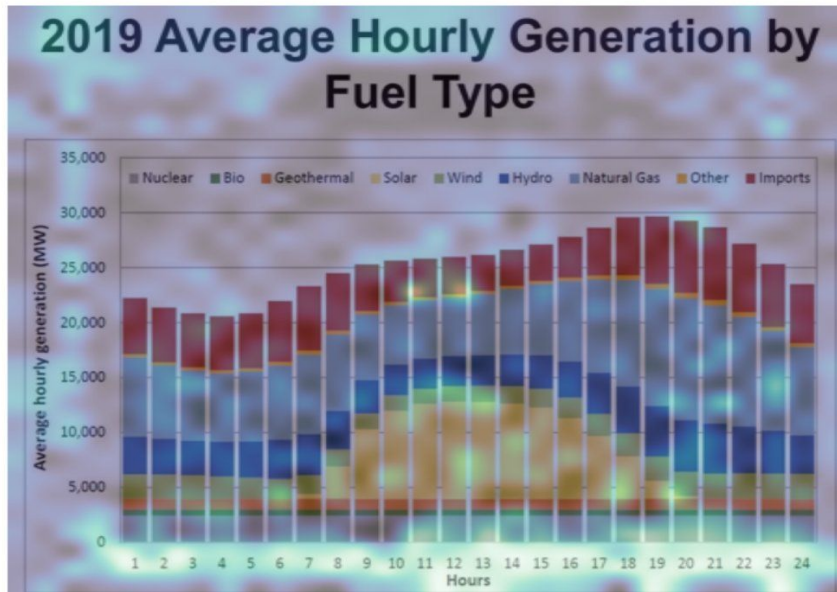


**USING COMPLEX RETRIEVAL SYSTEMS
THAT RELY ON OCR, DOCUMENT LAYOUT
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FIGURE CAPTIONING AND POWERFUL
TEXT EMBEDDING MODELS**



JUST EMBED THE IMAGE

ColPali



Query: "Which hour of the day had the highest overall eletricity generation in 2019?"

Figure 1: For each term in a user query, **ColPali** identifies the most relevant document image patches (highlighted zones) and computes a query-to-page matching score. We can then swiftly retrieve the most relevant documents from a large pre-indexed corpus.

Multi-vector (ColPali) support: Weaviate 1.29

Illustration: Single-vector vs multi-vector embeddings

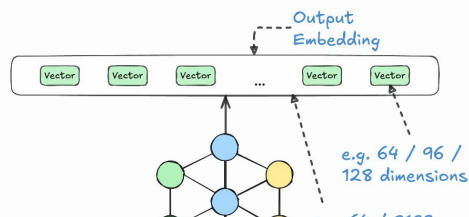
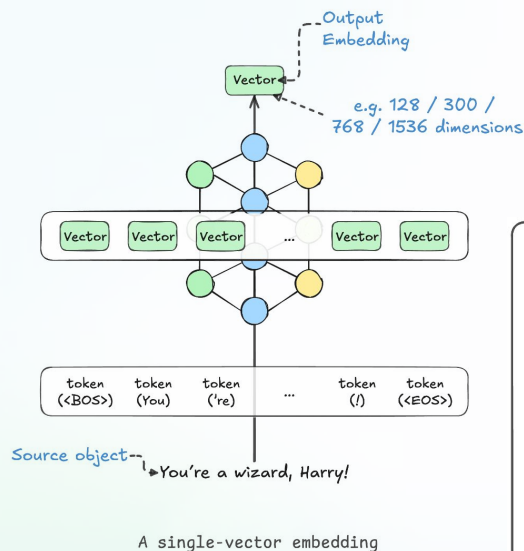
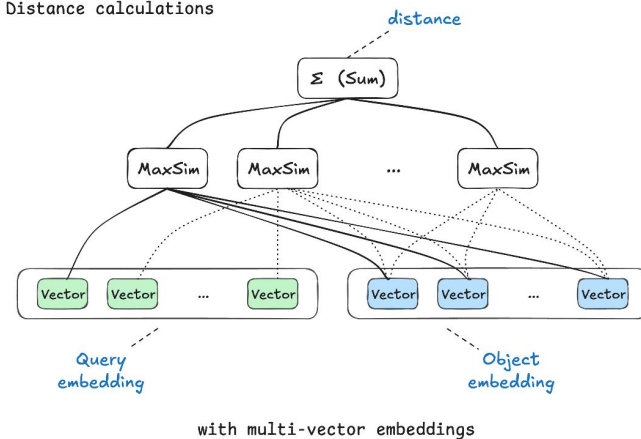
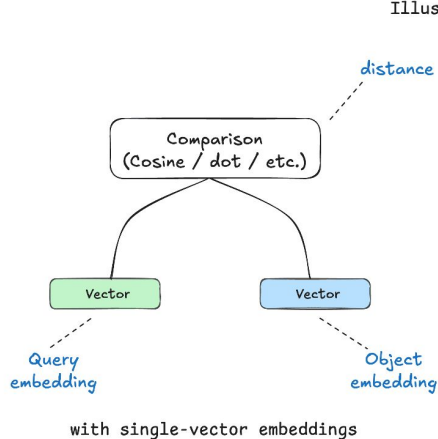


Illustration: Distance calculations





Want to learn more??

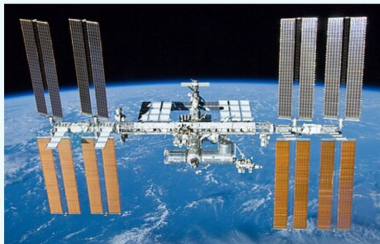
Weaviate Academy: Work with multimodal data

Image query

</> Code

This example finds entries in "MovieMM" based on their similarity to [this image of the International Space Station](#), and prints out the title and release year of the top 5 matches.

Query image



```
import weaviate
import weaviate.classes.query as wq
import os
```

Image query

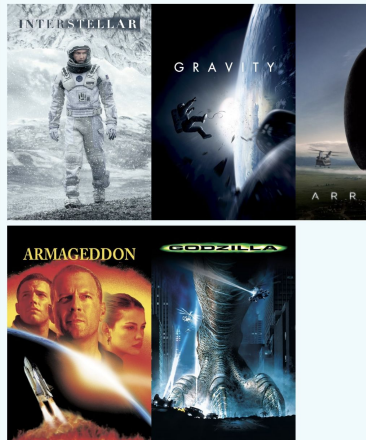
</> Code

Explain the code

Response object

Example results

Posters for the top 5 matches:



</> Code

This example finds entries in "MovieMM" with the highest keyword search scores for the term "history", and prints out the title and release year of the top 5 matches.

```
import weaviate
import weaviate.classes.query as wq
import os

# Instantiate your client (not shown), e.g.:
# headers = {"X-OpenAI-API-Key": os.getenv("OPENAI_APIKEY")} # Replace with your OpenAI API key
# client = weaviate.connect_to_local(headers=headers)

# Get the collection
movies = client.collections.get("MovieMM")

# Perform query
response = movies.query.bm25(
    query="history", limit=5, return_metadata=wq.MetadataQuery(score=True)
)

# Inspect the response
for o in response.objects:
    print(
        o.properties["title"], o.properties["release_date"].year
    ) # Print the title and release year (note the release date is a datetime object)
    print(
        f"BM25 score: {o.metadata.score:.3f}\n"
    ) # Print the BM25 score of the object from the query

client.close()
```

Keyword search

</> Code

Explain the code

Hybrid search

</> Code

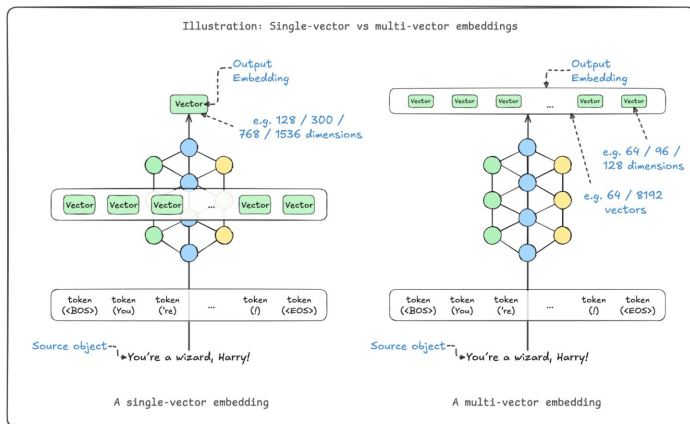
Explain the code

Questions and feedback

Want to learn more??

Weaviate Academy: Work with multimodal data

Weaviate docs: Multi-vector embedding tutorial



Multi-vector representations allow for more nuanced comparisons between objects, and therefore improved retrieval of similar objects.

Weaviate [1.29](#) introduces support for multi-vector embeddings, allowing you to store and search for objects using multi-vector embeddings.

Prerequisites

Introduction

When to use multi-vector embeddings

Option 1: ColBERT model integration

- 1.1. Connect to Weaviate
- 1.2. Collection configuration
- 1.3. Import data
- 1.4. Perform queries

Option 2: User-provided embeddings

- 2.1. Connect to Weaviate
- 2.2. Collection configuration
- 2.3. Import data
- 2.4. Perform queries

Summary

Further resources

Questions and feedback

IN DEPTH: UNDERSTANDING "LATE INTERACTION"

Late interaction is an approach for computing similarity between texts that preserves fine-grained meaning by comparing individual parts of the text (like words or phrases). Models like ColBERT use this technique to achieve more precise text matching than traditional single-vector methods.

The following visualization shows how late interaction works in a ColBERT model, in comparison to a single-vector model.

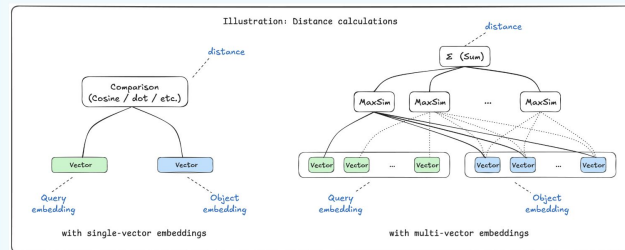


Figure: Late interaction vs single-vector comparison

More about late interaction

In a single-vector approach, two embeddings have the same dimensionality (e.g. 768). So, their similarity is calculated directly, e.g. by calculating their dot product, or cosine distance. In this case, the only interaction occurs when the two vectors are compared.

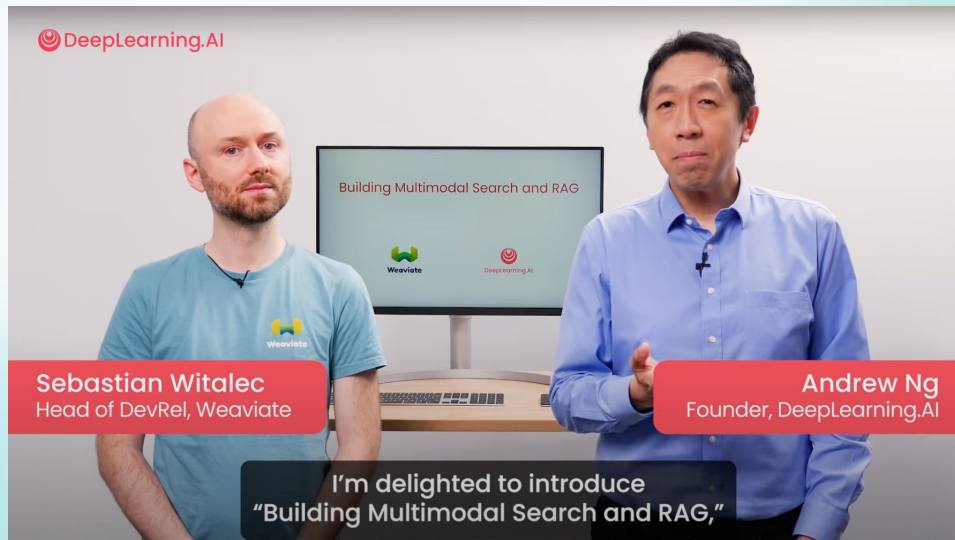


Want to learn more??

Weaviate Academy: Work with multimodal data

Weaviate docs: Multi-vector embedding tutorial

Deeplearning.ai: Multimodal search & RAG



Thank you!

